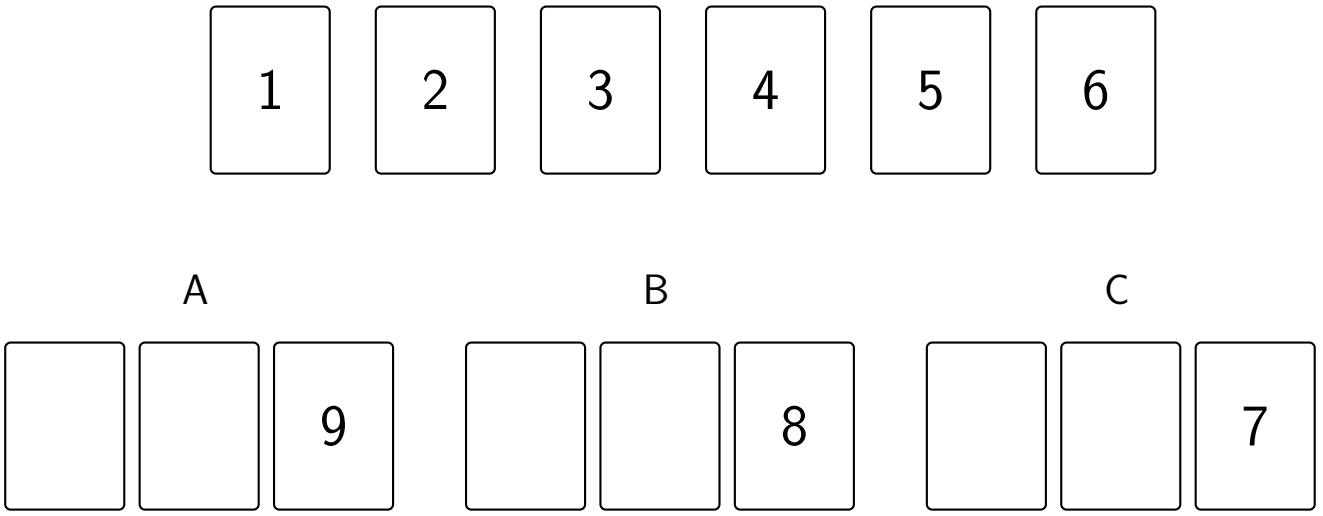
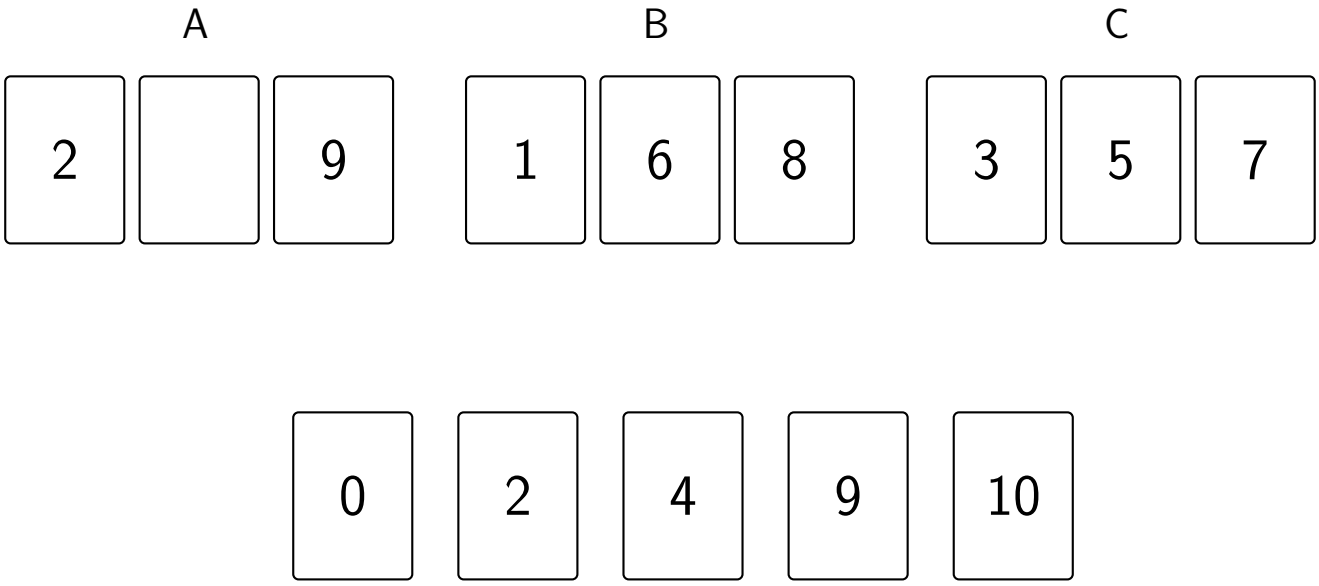
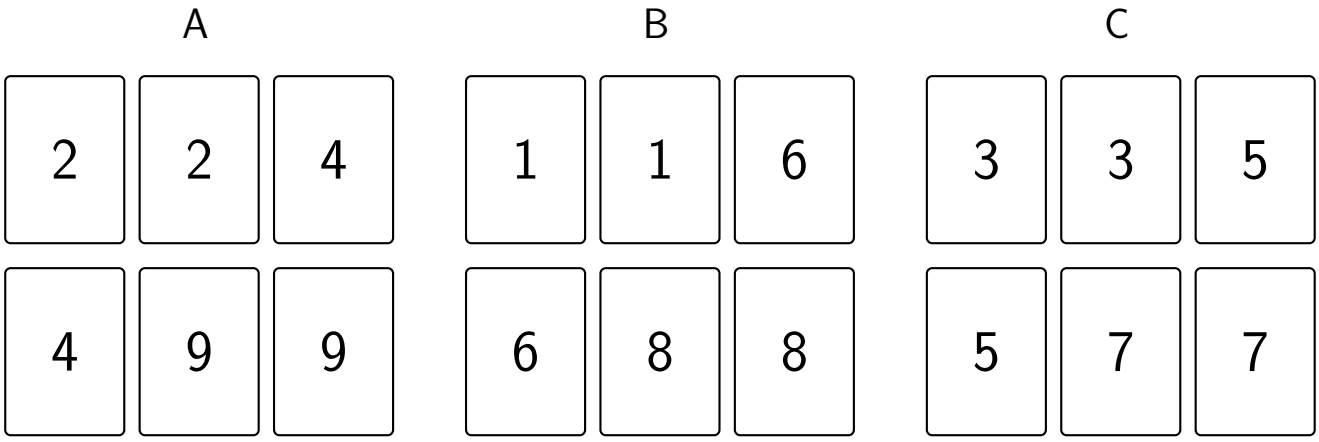
Problem 3: Keep 9 in A, 8 in B, and 7 in C. Place cards 1 to 6 so A wins more pairs than B, B more than C, and C more than A. Find two arrangements different from Problem 1.



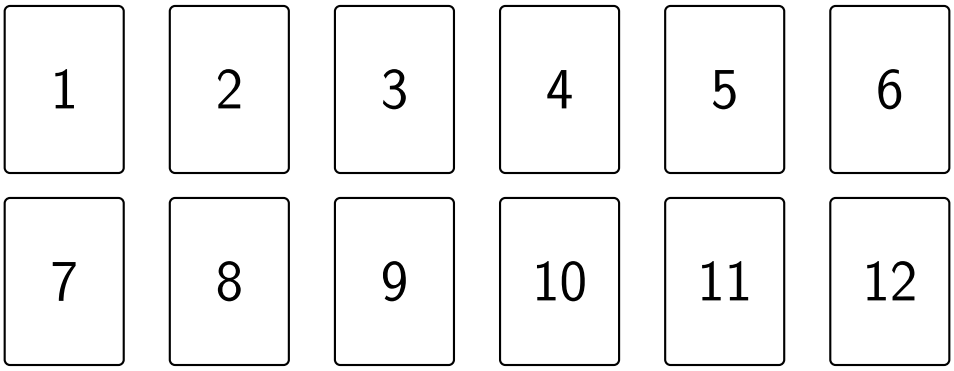
Problem 4: Choose 0, 2, 4, 9, or 10 for A's empty card. Find every choice that makes A win more pairs than B, B more than C, and C more than A.



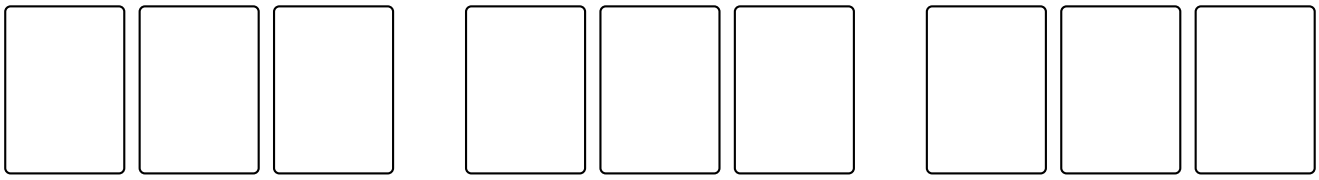
Problem 5: For each pair of these six-card decks, how many possible card pairs does each deck win? Does any winner change from Problem 1?



Problem 6: Choose nine different cards from 1 to 12 and make three decks of three. Make A win more pairs than B, B more than C, and C more than A, with totals 15 in A, 18 in B, and 21 in C.



A B C



Problem 7: Use cards 1 to 6 once each to make three two-card decks. Can A win more pairs than B, B more than C, and C more than A? Find an arrangement, or explain why there is none.

